Library Management

CryptoBooks: next-generation library management with NFTs and blockchain technology

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Abstract

Purpose – This study proposes CryptoBooks, a novel framework integrating cryptographic hashing and blockchain-based non-fungible tokens (NFTs) to ensure document authenticity, ownership and integrity in digital library environments.

Design/methodology/approach – A design science research approach is adopted to develop and demonstrate a prototype system combining SHA-512 hashing, InterPlanetary File System storage and Ethereum-based NFT minting. The workflow is implemented using Python scripting, decentralized storage and smart contract deployment via Remix Integrated Development Environment (IDE).

Findings – The prototype demonstrates that digital documents can be transformed into immutable, verifiable assets with embedded cryptographic fingerprints and decentralized ownership records. The system ensures tamper-evident verification, transparent provenance tracking and independent authentication without reliance on centralized authorities. Mapping the framework against established authenticity criteria shows that its three components correspond to the content, agent and time of creation of a record.

Practical implications – The framework supports digital preservation, forgery prevention and verifiable credentialing in academic libraries. Seven applications are identified, including provenance for special collections, priority claims in academic integrity disputes and mitigation of reference rot. A staged adoption roadmap is proposed whose first stage requires no distributed ledger and no expenditure.

Originality/value – While cryptographic hashing combined with token-based ownership is established in other sectors, this study is among the first to adapt the approach to document-level authentication in libraries and to map its components onto the conditions of authenticity codified in archival science.

Keywords CryptoBooks, Document authentication, Digital archives, Library blockchain, Library innovation, NFTs, Ownership, Library management

Paper type Conceptual paper

1. Introduction

Libraries have long been revered as the custodians of human civilization, spaces where knowledge is preserved, protected and made accessible across generations. In their traditional form, libraries safeguarded physical manuscripts, legal records, personal archives and cultural artifacts. With the digital transformation of society, these institutions have embraced digitization to expand access and preserve fragile materials. However, this shift has introduced a new and critical vulnerability: the authenticity and permanence of digital documents.

Digital files can be easily copied, edited or forged. Unlike physical records, digital documents do not always carry visible marks of originality. Over time, this leads to serious challenges: How can a library prove that a document has not been altered? How can ownership or authorship be verified years later? In a world where misinformation, data manipulation and digital forgery are rising, libraries must evolve from being passive information holders to active guarantors of truth and integrity.

Emerging technologies such as blockchain, Secure Hash Algorithm (SHA) hashing and non-fungible tokens (NFTs) offer powerful tools to meet this challenge. Together, they form

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the foundation of a new concept, CryptoBooks. A CryptoBook is a digital representation of a document that combines two core technologies:

(1) SHA-512 hashing, which generates a unique, tamper-proof fingerprint of the file; and

(2) NFTs, which register the document as a unique, time-stamped asset on a blockchain, tied to a verifiable owner.

When applied within library systems, CryptoBooks ensure that any document, whether a property deed, student thesis, rare map or birth certificate, can be verified for authenticity, originality and ownership, even decades after it was created. This innovation holds profound implications: it empowers users to protect their personal and legal records, it enables libraries to combat digital fraud and it opens a new chapter in archival science.

This research paper investigates how libraries can integrate CryptoBooks into their digital services. It explores the technical workflow, benefits and challenges of implementing SHA-512-based document verification and NFT-backed ownership within academic and public libraries. By doing so, it proposes a model where libraries become not only digital repositories but fortresses of trust, where truth is recorded, not just stored, and where every document is sealed with proof.

Despite the growing body of literature on blockchain and NFTs in library systems, existing studies remain fragmented, primarily focusing on isolated aspects such as blockchain infrastructure or metadata management. There is a clear lack of integrated frameworks that combine cryptographic hashing with NFT-based ownership for document-level authentication in libraries. Addressing this gap, this study proposes and evaluates the *CryptoBooks* framework, which integrates cryptographic hashing, decentralized storage and blockchain-based ownership mechanisms into a unified and implementable model for secure digital document verification.

Accordingly, the study pursues the following objectives:

(1) To design a CryptoBook-based document authentication framework

(2) To integrate SHA-512 hashing with NFT-based ownership

(3) To demonstrate feasibility using a prototype implementation

(4) To evaluate applicability in library environments

2. Literature review

The evolution of library services from traditional print-based models to digital infrastructures has introduced significant challenges concerning document authenticity, ownership verification and long-term digital preservation. Emerging technologies such as blockchain, cryptographic hashing and NFTs offer innovative tools to address these issues. A comprehensive review of relevant literature reveals growing academic interest in these technologies and their intersection with library science.

2.1 Applications of blockchain and NFTs in other sectors

The pairing of cryptographic hashing with blockchain-registered tokens is not unique to libraries. It already operates at considerable scale across several industries in which authenticity, provenance and ownership carry commercial or legal weight, and the experience accumulated in those settings offers valuable evidence of its viability.

In the luxury goods sector, the Aura Blockchain Consortium was founded in 2021 by LVMH Moët Hennessy Louis Vuitton SE (LVMH), Prada Group, Cartier and Only The Brave (OTB) Group to combat counterfeiting. Each product is registered with a unique digital identity on a permissioned ledger, allowing purchasers to trace an item through its supply chain

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and verify its authenticity. By 2025, the consortium reported more than fifty member brands and over sixty million registered products (Aura Blockchain Consortium, 2025). Two aspects of this deployment are instructive for libraries. The consortium adopted a permissioned rather than a public ledger, which removes transaction fees and keeps governance with the participating institutions, and competing firms cooperated on shared infrastructure because authenticity is a collective rather than a proprietary benefit.

In healthcare, Azaria *et al.* (2016) developed MedRec, a decentralized record management system in which the ledger stores pointers and access permissions while the clinical record itself remains in the provider system. This separation of the authenticity layer from the content layer is the same architectural choice adopted in this study, where only the hash and the storage identifier are written to the chain. MedRec also demonstrated that a hash-linked audit trail makes attempted manipulation immediately visible across the network.

In education and credentialing, the Blockcerts standard developed by the Massachusetts Institute of Technology issues tamper-evident digital diplomas. A one-way hash of the credential is written to the blockchain, while the credential itself is delivered to the graduate as a file bound to their private key (MIT News, 2017). Massachusetts Institute of Technology (MIT) progressed from a pilot cohort in 2017 to institution-wide issuance, and other universities have since adopted the standard. Pu and Lam (2023) analyze the benefits realized across several such deployments, identifying reduced verification costs and improved decision making among the common outcomes. This is the closest existing analog to the framework proposed here, and it is significant that it emerged from a registrar function rather than a technology unit.

In digital rights management, Frattolillo (2024) presents a blockchain- and smart-contract-based watermarking protocol for digital copyright protection that dispenses with the trusted third party on which earlier schemes relied. The relevance here is the underlying principle: immutable registration of a work at a given moment establishes a defensible claim without requiring an intermediary to vouch for it.

These precedents establish that the approach is both technically and commercially viable. What distinguishes the library case is not the cryptography itself but the requirements that surround it. Libraries operate on preservation horizons measured in decades or centuries rather than in product lifecycles, they are bound to established metadata and persistent identifier standards, and their holdings are frequently subject to third-party rights and privacy constraints. The contribution of this study therefore lies in adapting a demonstrated pattern to these distinctive conditions.

2.2 Blockchain and NFTs in library and information science

Jha (2023) emphasized the importance of blockchain integration in libraries to enhance document authentication, user privacy and provenance management. He underscored the synergy of cryptographic hashing and NFTs in ensuring secure digital records within decentralized environments. Extending this, Suman and Patel (2021) explored how blockchain technology could strengthen core library services, particularly metadata management, decentralized cataloging and interlibrary transactions. Their conceptual framework aligns closely with the CryptoBooks approach by advocating blockchain immutability and transparency for archival integrity.

Further substantiating these claims, Usman and Ullah (2024) examined blockchain implications for user satisfaction, operational efficiency and data protection in libraries. They concluded that the decentralized and tamper-proof nature of blockchain enhances the trustworthiness and traceability of library services. Their study offered a practical blueprint for implementing blockchain-driven modules to reduce manual intervention and improve information flow.

In parallel, Delgado-von-Eitzen *et al.* (2024) presented a GDPR-compliant model that utilizes NFTs for issuing academic credentials. Their work focused on giving individuals granular control over their digital records while ensuring legal and ethical compliance, thus offering a framework highly compatible with CryptoBooks, especially in academic

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repositories. Putnings (2022) contributed to this discourse by advocating the use of NFTs in scholarly publishing. He emphasized the significance of digital uniqueness and ownership verification, noting that NFTs can restore the lost value of scientific artifacts in the open-access digital ecosystem.

Complementing the blockchain narrative, cryptographic techniques have also been explored for their potential in document verification. Sumagita and Riadi (2018) conducted a comparative study of cryptographic hash functions and concluded that SHA-512 is superior in providing resistance against collision attacks, making it particularly useful in secure login systems and digital integrity checks. This reinforces the adoption of SHA-512 by CryptoBooks for generating verifiable document fingerprints.

The role of blockchain in academic transparency and copyright assurance has also been studied extensively. Chen *et al.* (2018) argued that blockchain could be utilized in academia to track intellectual property, protect copyrights and ensure authorial legitimacy, particularly through smart contracts and decentralized metadata storage. Hoy (2017) highlighted the capacity of blockchain to generate tamper-evident, timestamped versions of digital records, proposing its use for both rights management and publication verification.

Tenorio-Fornés *et al.* (2019) envisioned a decentralized publishing ecosystem where blockchain enables peer-review reputation tracking, transparent governance and secure document workflows. Their proposed infrastructure illustrates how blockchain could address long-standing issues of fairness and accessibility in academic publishing. Similarly, Gul and Bano (2019), as reported by Suman and Patel (2021), describe Orvium, a blockchain-based publishing platform that compensates authors using token systems and automates peer review. These platforms exemplify the value proposition that CryptoBooks can deliver within library-linked publishing networks.

Practical library implementations of blockchain are emerging globally. Oyelude (2019) reviewed trends in this area, including LibChain, a blockchain-based distributed library management system that allows users to loan books directly to peers and supports interlibrary lending without central mediation.

Dubey and Verma (2024) introduce the concept of *Crypto Books*, proposing the use of NFTs as a mechanism to certify ownership and provenance of both physical and digital assets within library environments. They argue that by leveraging the immutable ledger of blockchain and the unique, tamper-evident nature of NFTs, libraries can enhance digital preservation, streamline access control and bolster overall security. The authors outline the technical architecture underpinning Crypto Books, detailing the minting process of NFTs and their integration into existing library management systems (LMS). They also explore key implementation issues, including costs, privacy and technical complexity, while identifying future avenues for research and development. This study builds directly on that conceptual foundation, advancing it from proposition to implemented and evaluated prototype.

Collectively, these studies establish a strong theoretical and practical foundation for the CryptoBooks model, which synthesizes SHA-512 hashing, NFT metadata embedding and blockchain-based ownership registration. The integration of these technologies not only ensures tamper-resistance and authenticity but also democratizes document verification by removing reliance on central authorities. While technical, legal and operational challenges remain, the literature affirms that CryptoBooks can significantly enhance trust, transparency and preservation in digital library ecosystems.

2.3 Theoretical framework: authenticity in archival science

Archival science distinguishes authenticity from reliability and accuracy. International Organization for Standardization (ISO) 15489–1 defines an authentic record as one that can be proven to be what it purports to be, to have been created or sent by the person purported to have created or sent it and to have been created or sent at the time purported (International

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Organization for Standardization, 2016). These three conditions, which may be summarized as content, agent and time, provide a standard against which any authentication mechanism can be evaluated.

The InterPARES project elaborated this standard into benchmark requirements supporting the presumption of authenticity and baseline requirements supporting the certification of copies by an archival institution (MacNeil *et al.*, 2001; Duranti and Preston, 2008). A central finding of that program is that authenticity in digital environments cannot be inferred from the object alone, since digital objects lack the physical indicia on which traditional diplomatics relied. It must instead be established through the recordkeeping system and maintained across successive custodial transfers. The records continuum model (Upward, 1996, 1997; McKemmish, 2001) adds that record creation, capture and organization are coexisting dimensions rather than sequential stages, which implies that authenticity controls are most effective when applied at the moment of deposit rather than retrofitted at archival transfer.

Assessed against these criteria, the three components of the CryptoBooks framework correspond directly to the three conditions of authenticity, as set out in Table 1.

Table 1. Mapping of CryptoBooks components to the ISO 15489 conditions of authenticity

Condition of authenticity	Archival question	CryptoBooks component	Evidence produced
The record is what it purports to be	Has the content been altered?	SHA-512 hash	Recomputed hash matches the stored value
Created or sent by the person purported	Who deposited the record?	NFT ownership record	Depositor wallet address recorded on chain
Created or sent at the time purported	When was it deposited?	Block timestamp	Immutable time of the minting transaction

This correspondence grounds the CryptoBooks framework in a body of archival scholarship that considerably predates the technologies it employs. Far from representing a departure from established practice, the framework offers a computational means of satisfying requirements that archivists have long recognized as fundamental to the trustworthiness of records. It is equally important to recognize what the framework does not claim. A CryptoBook demonstrates that a record has remained unchanged since the moment of deposit and that a specific wallet key authorized that deposit. It does not, by itself, establish authorship, originality or the right to deposit, since the binding of a person to a key remains a question of identity that libraries continue to address through their existing authentication systems.

Questions of institutional uptake are approached through the diffusion of innovations framework (Rogers, 2003), which identifies relative advantage, compatibility, complexity, trialability and observability as the attributes that govern how rapidly an innovation spreads through a professional community. These attributes inform the adoption roadmap developed later in this study, where the earliest stage is deliberately designed to be simple to attempt and simple to reverse.

2.4 Conceptual framework

2.4.1 What is CryptoBooks?. CryptoBooks are cryptographically secured digital records that combine NFTs and cryptographic hash functions (e.g. SHA-512) to ensure authenticity, ownership and tamper-proof archiving of digital documents in library environments. At their core, CryptoBooks serve as digital certificates of authenticity, permanently recorded on a blockchain. Each CryptoBook represents a unique document, such as a thesis, property record,

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research article or rare manuscript, whose integrity and ownership can be verified without relying on any centralized authority.

2.4.2 Integration of NFTs and SHA hashing in library systems. The integration of NFTs and SHA hashing into library systems offers a modern, secure way to manage digital documents, authenticate ownership and preserve long-term integrity. This system works by first applying a SHA such as SHA-512 to a digital document (e.g. thesis, manuscript or institutional archive). This produces a unique hash value that serves as a tamper-evident fingerprint of the document.

Once the hash is generated, it is embedded into the metadata of an NFT, which is then minted on a blockchain platform like Ethereum. This NFT acts as a digital certificate of authenticity and ownership, linking the document to its rightful creator or custodian. The NFT metadata may also include information like document title, creation date, creator name and a link to the file stored on decentralized platforms such as InterPlanetary File System (IPFS).

Libraries can integrate this system into their institutional repositories, digital archives or LMS using APIs or plugins that connect blockchain platforms to the library backend. When a user submits a document, the system automatically generates a hash, mints the corresponding NFT and records the data. Later, anyone can verify the authenticity of the document by re-hashing it and comparing the output with the NFT record on the blockchain.

This integration enables libraries to

(1) Ensure document authenticity and prevent tampering.

(2) Provide verifiable proof of ownership.

(3) Support secure digital archiving for long-term preservation.

(4) Build a decentralized trust infrastructure for scholarly communication.

(5) Such systems are especially useful in academic, research and archival libraries where document integrity and provenance are critical.

2.4.3 Security model: ensuring ownership and integrity. The security model underpinning the CryptoBooks framework relies on two fundamental principles: document ownership and content integrity. By integrating NFTs with cryptographic hash functions such as SHA-512, libraries can implement a tamper-resistant and verifiable digital record system. This model guarantees that each document can be uniquely attributed to its rightful owner and that its contents remain unaltered over time.

Ownership via NFTs: Ownership is authenticated through the creation of a blockchain-based NFT, which serves as a unique digital certificate for each document. When a document is archived, an NFT is minted containing key metadata, including the cryptographic hash of the document, the blockchain wallet address of the owner or creator, a timestamp marking the registration of the document and optional metadata such as title, description and a reference link to a decentralized file storage system like IPFS. These NFTs are immutable and transparent on the blockchain, allowing any third party to verify the ownership and provenance of a document without relying on a central authority. This is particularly useful in library settings where authorship, licensing and provenance are critical.

Integrity via SHA hashing: To ensure the integrity of a digital document, it is hashed using a secure cryptographic algorithm such as SHA-512. This process generates a unique and fixed-length hash value that acts as a digital fingerprint for the file. Even a minor alteration in the document will result in a significantly different hash, thereby signaling tampering. When a user or system wishes to verify the document, the file is re-hashed, and the result is compared with the hash value stored in the NFT. If both values match, the document is confirmed to be authentic and unmodified.

Combined protection: By leveraging both NFTs and SHA hashing, the CryptoBooks model establishes a two-tier security mechanism in which NFTs ensure document ownership and traceable provenance, protecting intellectual property and authorship rights, while SHA hashes guarantee the integrity of document contents, safeguarding against unauthorized edits

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or corruption. This model provides a decentralized, transparent and durable framework suitable for use in academic libraries, institutional repositories and digital archives. It eliminates the need for centralized verification systems and builds trust among users, researchers and custodians of knowledge.

2.5 Research gap

Despite the growing body of research on blockchain applications and NFTs in libraries, existing studies remain fragmented, focusing primarily on individual components such as metadata management, credentialing or decentralized storage. There is a notable lack of integrated frameworks that combine cryptographic hashing, NFT-based ownership and decentralized storage into a unified, implementable model for document-level authentication. This limitation highlights the need for a comprehensive approach, which this study addresses through the CryptoBooks framework.

3. Methodology

This study adopts a design science research approach (Hevner *et al.*, 2004; Peffers *et al.*, 2007), focusing on the development and evaluation of a prototype system integrating blockchain and cryptographic techniques for library applications. The methodological framework involves the design, development and implementation of a secure digital verification model that integrates SHA-512 cryptographic hashing, blockchain-based NFT and decentralized storage using the IPFS.

The prototype system was evaluated based on the following criteria: (1) accuracy and consistency of hash generation during document verification, (2) successful NFT minting and correct metadata linkage, (3) reliability of document storage and retrieval via IPFS content identifiers (CID) and (4) integrity validation through hash comparison. These evaluation parameters ensure that the system meets the requirements of authenticity, ownership verification and tamper-resistance in digital library environments.

3.1 Design of the CryptoBook system

The CryptoBook system is conceptualized as a decentralized digital authentication framework that facilitates secure ownership and integrity validation of documents within library systems. It is built on a modular design, integrating cryptographic hashing, blockchain-based NFT minting and decentralized storage mechanisms. This system ensures that each digital document archived or accessed through a library can be authenticated for originality, attributed to its rightful owner and verified for any alterations over time. The system is designed to operate independently of centralized authorities, promoting trustless verification and long-term digital preservation.

3.2 Tools and platforms used

Table 2 summarizes the tools and platforms employed.

3.3 Coding environment

(1) Smart contract layer (solidity): Ethereum smart contracts are written in solidity using the ERC-721 standard to mint NFTs that represent unique documents.

(2) Hashing and verification layer (Python): Python scripts are used to compute cryptographic hashes (SHA-512) of documents and compare these hashes during verification.

(3) Decentralized storage (IPFS Command Line Interface (CLI)/Application Programming Interface (API)): Document files or their hash values are uploaded to IPFS, returning a unique CID, which is stored in the metadata of the NFT.

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Table 2. Tools and technologies used in the CryptoBook prototype

Tool/Platform	Purpose in CryptoBook system
Python (hashlib library)	Used to generate the SHA-512 cryptographic hash of the digital document to create a unique fingerprint for integrity verification
SHA-512 hash algorithm	Provides a secure cryptographic digest ensuring that any modification in the document can be detected
IPFS (InterPlanetary File System)	Decentralized storage network used to store the document and metadata files securely and immutably
Pinata IPFS gateway	Cloud-based IPFS pinning service used to upload and persist document files and NFT metadata on the IPFS network
JSON metadata format	Structured data format used to store document attributes such as hash value, CID and descriptive metadata for the NFT
Solidity programming language	Used to develop the ERC-721 smart contract responsible for NFT minting and metadata linking
Remix Ethereum IDE	Browser-based development environment used for compiling, deploying and testing the CryptoBookNFT smart contract
ERC-721 token standard	Ethereum standard used to represent each digital document as a unique non-fungible token
Ethereum Virtual Machine (Remix VM)	Simulated blockchain environment used for testing and executing smart contract deployment and NFT minting operations

3.4 Steps to create a CryptoBook

The creation and verification of a CryptoBook involve four key steps, presented in Figure 1.

(1) **Document selection and hashing**: A digital document (e.g. thesis, manuscript, legal record) is selected and hashed using SHA-512 via a Python script. The resulting hash serves as a unique, tamper-evident identifier.

(2) **Decentralized storage using IPFS**: The document or its hash file is uploaded to the IPFS, which generates a CID, a permanent, decentralized storage link.

(3) **NFT minting with metadata**: A smart contract on the Ethereum blockchain is used to mint an NFT. The NFT metadata includes the document hash, the wallet address of the owner, a timestamp, the IPFS CID, title and description.

(4) **Ownership and verification**: The NFT acts as proof of ownership and authenticity. Any user can later retrieve the document, recompute the hash and compare it with the hash stored in the NFT to verify integrity and originality.

3.5 Scope and delimitation

This study is limited to demonstrating the technical feasibility of CryptoBooks within digital libraries. It does not include legal policy validation, real-world user deployment or large-scale institutional implementation. However, it provides a blueprint that can be adopted by academic and public libraries for pilot projects. It should be noted that contract deployment and minting were carried out on the Remix virtual machine, a simulated Ethereum environment, and that no transaction was submitted to a public network. The prototype therefore establishes the correctness of the contract while leaving questions of transaction cost and confirmation latency to be examined separately, as discussed in Section 6. The complete workflow is represented in Figure 2.

4. Prototype implementation

This section presents the prototype implementation of the CryptoBook system using a Python-based approach, focusing on digital document hashing, decentralized storage integration and

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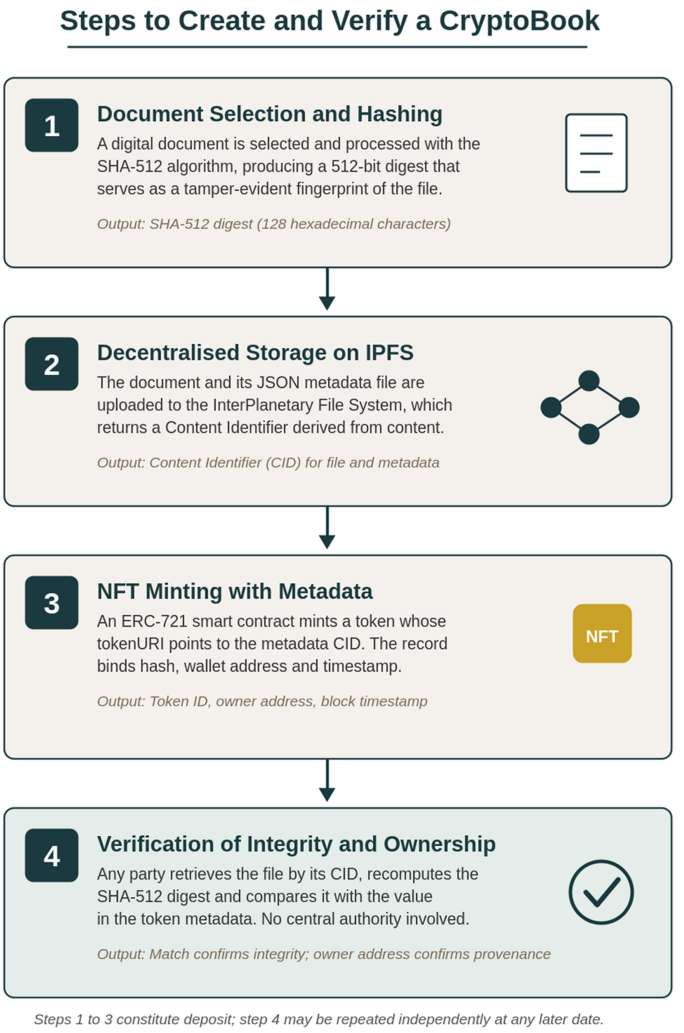


Figure 1. Steps to create and verify a CryptoBook

metadata generation. The goal is to demonstrate a practical, accessible and secure workflow that libraries can adopt for document authentication and preservation.

4.1 Document hashing using Python (SHA-512)

A SHA-512 hash of the digital document is generated first, functioning as a unique and irreversible fingerprint of the file. A Python script using the hashlib library was applied to a sample document, *thesis.pdf*, reading the file in fixed-size chunks so that memory use remains independent of document size. The script and its output are shown in Figure 3 and the resulting parameters in Table 3.

Table 3 demonstrates that the SHA-512 algorithm generates a fixed-length cryptographic hash, ensuring strong collision resistance and reliable document fingerprinting for integrity verification.

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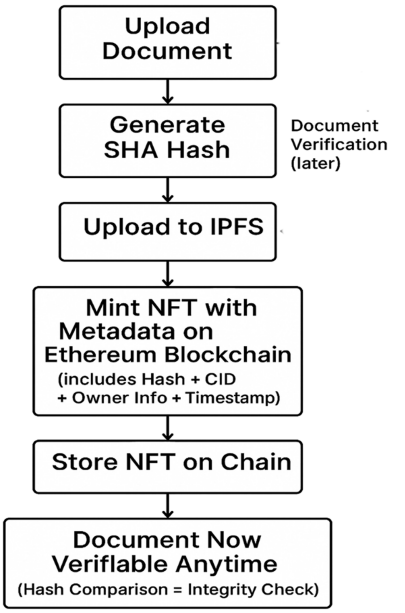


Figure 2. Process flowchart

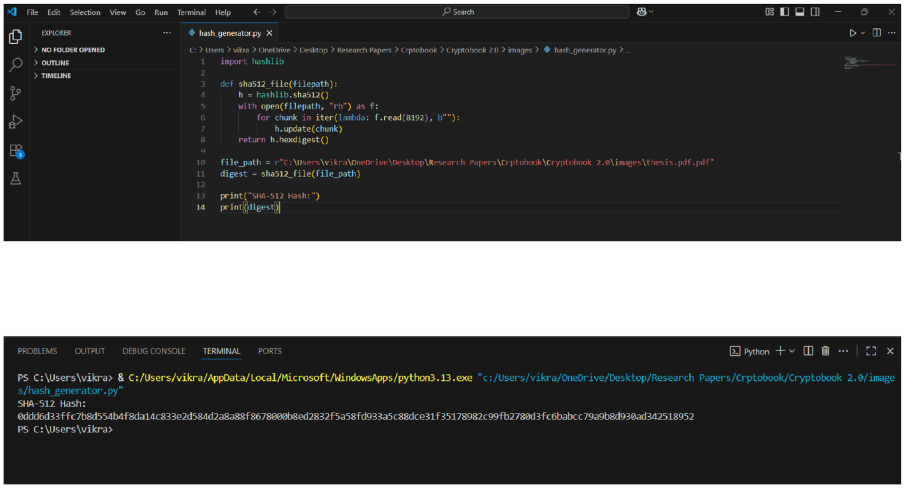


Figure 3. SHA-512 hash generated for the sample document using Python (input code above, output below)

4.2 Uploading to IPFS for decentralized storage

The sample document was uploaded to IPFS using the Pinata gateway (Figure 4). IPFS stores files using a content-addressed mechanism in which each file is assigned a CID, allowing retrieval from any IPFS gateway without reliance on a centralized server. In Table 4, it shows that the CID is later embedded into the NFT metadata to enable decentralized verification and retrieval.

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Table 3. SHA-512 hash generation and document integrity parameters

Parameter	Value
Hash algorithm	SHA-512
Generated hash	0ddd6d33ffc7b8d554b4f8da14c833e2d584d2a8a88f8678000b8ed2832f5a58fd933a5c88d ce31f35178982c99fb2780d3fc6babcc79a9b8d930ad342518952
Hash length	512 bits
Purpose	Document fingerprint for NFT-linked verification

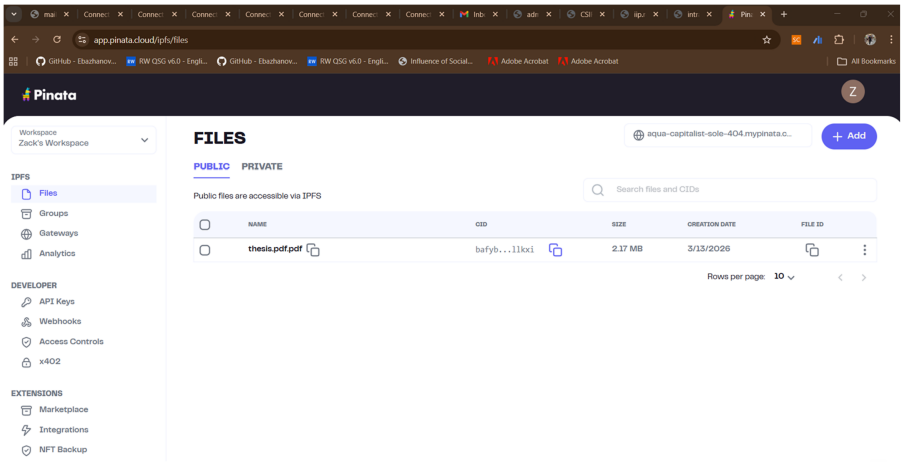


Figure 4. Document uploaded to IPFS

Table 4. IPFS storage and document retrieval details

Parameter	Value
Storage system	InterPlanetary File System (IPFS)
Upload platform	Pinata
File name	thesis.pdf
File size	2.17 MB
Generated CID	bafybeieshh2szrak4qued3ahezroqnbu2jp24z62ejwe5i4lh4jgfllkxi
Retrieval gateway	https://gateway.pinata.cloud/ipfs/ bafybeieshh2szrak4qued3ahezroqnbu2jp24z62ejwe5i4lh4jgfllkxi

4.3 Metadata generation

A metadata file in JSON format was created containing the document name, IPFS CID, cryptographic hash value and provenance information. This metadata serves as the digital descriptor of the CryptoBook record and is referenced during NFT minting. The file was itself uploaded to IPFS, generating a second CID which acts as the tokenURI reference for the NFT (Figure 5, Table 5).

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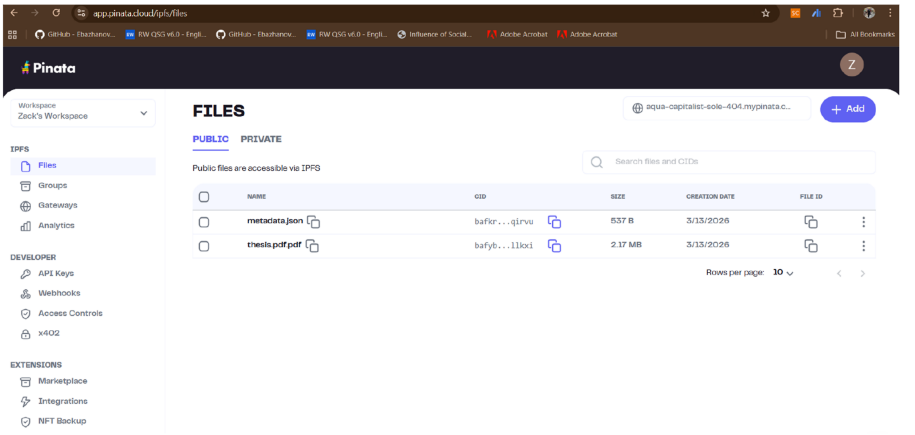


Figure 5. Metadata JSON uploaded

Table 5. NFT metadata storage and CID details

Parameter	Value
Metadata file	metadata.json
Storage platform	IPFS (Pinata)
Metadata CID	bafkreibboulspz2tiasyvkrokw7r4ppwrgull3g7hrmxpjqda2u2wqirvu
Purpose	NFT tokenURI reference

{
“name”: “CryptoBook Record #1”,
“document_name”: “thesis.pdf”,
“file_cid”: “bafybeieshh2szrak4qued3ahezroqnbu2jp24z62ejwe5i4lh4jgfllkxi”,
“hash_algorithm”: “SHA-512”,
“document_hash”: “0ddd6d33ffc7b8d554b4f8da14c833e2d584d2a8a88f8678000b8ed
2832f5a58fd933a5c88dce31f35178982c99fb2780d3fc6babcc79a
9b8d930ad342518952”
}

4.4 Smart contract implementation

A prototype ERC-721 smart contract, CryptoBookNFT, was developed, compiled and deployed using the Remix Ethereum IDE, as shown in Figures 6 and 7. The contract inherits the ERC721URIStorage extension and exposes a mint function accepting a recipient address and a tokenURI.

4.5 NFT minting and blockchain registration

During minting, the metadata CID stored on IPFS was provided as the tokenURI parameter of the NFT, establishing a permanent blockchain link between the token and the verified

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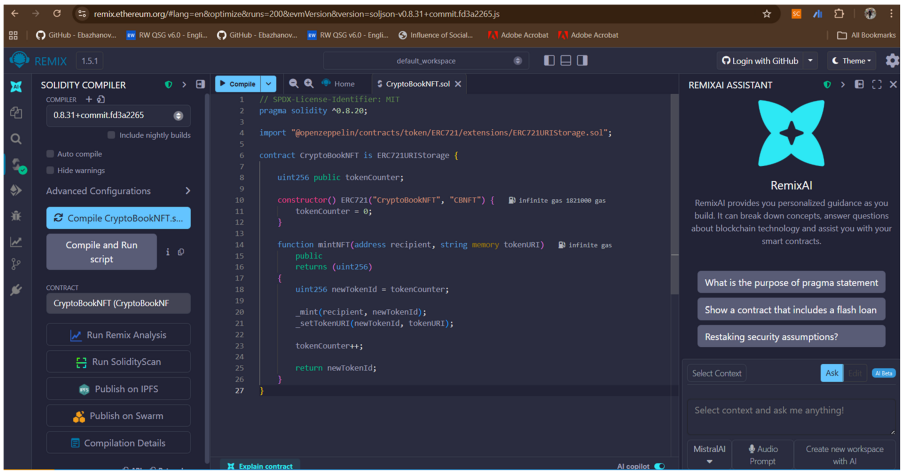


Figure 6. Compilation of the CryptoBookNFT smart contract using Remix Ethereum IDE

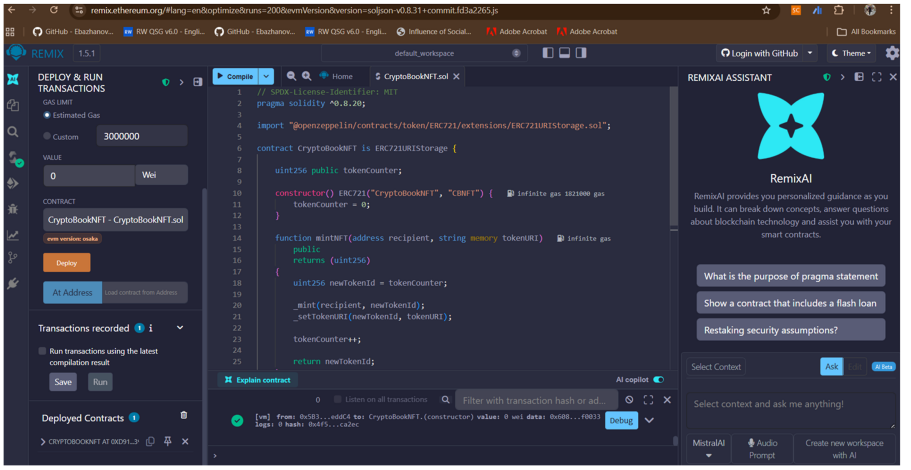


Figure 7. Deployment of the CryptoBook NFT smart contract in the Remix virtual blockchain environment

document record. Successful execution of the mintNFT function, shown in Figure 8, confirms the feasibility of integrating cryptographic hashing, decentralized storage and blockchain-based provenance tracking within the CryptoBook framework.

Table 6 highlights the successful deployment of the smart contract and NFT minting process, establishing a verifiable connection between the document, metadata and blockchain ownership record. The prototype demonstrates that each digital document can be uniquely represented as a blockchain token whose metadata references the decentralized IPFS storage location and the cryptographic hash of the document.

4.6 Discussion and validation

The findings of this study align with previous research on blockchain-based document authentication (Chen *et al.*, 2018; Delgado-von-Eitzen *et al.*, 2024), while extending existing

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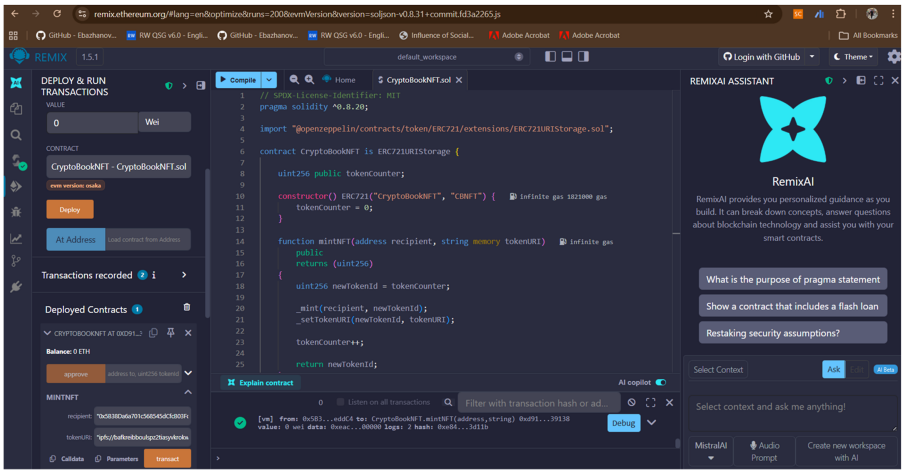


Figure 8. Execution of the mint NFT function in the Remix Ethereum IDE, demonstrating successful minting of the CryptoBook NFT linking the IPFS metadata to the blockchain token

Table 6. Blockchain and NFT configuration of the CryptoBook system

Parameter	Value
Blockchain environment	Remix VM (Ethereum Virtual Machine)
Smart contract	CryptoBookNFT
Token standard	ERC-721
Token ID	0
Metad ata URI	ipfs://bafkreibboulspz2tiasyvkrokw7r4ppwrgull3g7hrmxpjqda2u2wqirvu
Storage system	IPFS (Pinata)

work by integrating cryptographic hashing with NFT-based ownership into a unified library-oriented framework. The prototype implementation supports these claims by demonstrating consistent hash verification, successful NFT minting and reliable document retrieval through IPFS.

Integrity verification in CryptoBooks operates on the principle of deterministic hashing. What makes this process significant in a library context is its ability to eliminate ambiguity across decentralized storage and distributed access points. When a user retrieves a document from IPFS or another decentralized archive, they do not require access to any library server or administrator. A simple hash comparison confirms whether the file has remained unchanged since it was sealed in the CryptoBook. This user-driven model democratizes verification, allowing any stakeholder, researcher, auditor, publisher or librarian to validate authenticity with full autonomy.

In terms of ownership validation, CryptoBooks leverage blockchain transparency to maintain traceability. Each NFT minted within the system captures the provenance of a document, recording the blockchain address of the original depositor. Ownership is publicly verifiable, but not editable, making the record legally and ethically robust. Ownership verification is not passive: it can be triggered dynamically during user queries, rights disputes, archival audits or citation assessments. Compared to traditional centralized repository systems, which rely on institutional trust and administrative control, the CryptoBooks

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framework introduces a decentralized verification mechanism, reducing dependency on intermediaries and enhancing transparency.

It is important, however, to state the scope of this evidence with precision. The blockchain record establishes that a particular wallet key authorized a deposit at a particular moment; it does not, on its own, confirm that the holder of that key is the author of the work. In practice, libraries would resolve this by binding wallet addresses to verified institutional identities at the point of deposit, drawing on the authentication systems they already operate for repository submission.

A key strength of the CryptoBooks framework lies in its approach to dispute resolution, especially concerning claims of false authorship or ownership. In conventional systems, these issues often result in lengthy administrative reviews. In a CryptoBook-enabled ecosystem, claims can be challenged or dismissed using a combination of metadata timestamping, hash validation and wallet address verification. These three factors create an immutable triangulation of identity, content and chronology. As a result, institutional libraries can offer real-time validation support for students, researchers and courts, something that traditional repositories cannot efficiently provide.

This demonstrates that the CryptoBooks framework not only aligns with existing blockchain-based approaches but also advances the field by offering an integrated and library-oriented solution for document-level authentication and ownership verification. By combining cryptographic security with decentralized transparency, the model provides libraries with a strategic mechanism to protect intellectual property, defend academic integrity and support knowledge sovereignty in the digital age.

5. Applications in libraries

The CryptoBook system offers powerful applications in libraries by securing digital archives, rare manuscripts and personal records with verifiable authenticity and ownership. Through blockchain-based NFTs and cryptographic hashing, libraries can ensure tamper-evident preservation, decentralized ownership tracking and non-repudiation of documents. The applications set out below illustrate where this capability delivers practical value, beginning with those a library may implement using its existing infrastructure and progressing toward those that require agreement across institutions.

5.1 Provenance and chain of custody for special collections

Special collections frequently carry acquisition histories whose evidentiary basis rests on paper files that are themselves difficult to authenticate. By sealing each digitized item together with its provenance documentation at the point of accession, a library creates a fixed and verifiable record of what it held and when it recorded holding it. Later additions to the provenance file are appended rather than substituted, so the sequence of claims remains open to inspection. In restitution and repatriation discussions, where the credibility of institutional records is often the central question, the ability to demonstrate that a record has not been revised after the fact carries considerable weight. The institutional cost is modest, since the mechanism applies to accession records rather than to entire collections.

5.2 Academic integrity and priority claims

To illustrate practical applicability, consider a scenario where a university student submits a thesis to the institutional repository. The document is hashed, uploaded to IPFS and minted into an NFT using the wallet address of the student. Years later, when applying for a patent or academic credit, the student or an external agency can retrieve the thesis via the IPFS CID, compute its hash and match it to the NFT metadata publicly stored on the blockchain. The original wallet address, recorded as the NFT owner, acts as digital proof of deposit. The

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timestamp provides a tamper-evident record of submission date, which is critical in resolving priority claims.

The same mechanism supports examination boards investigating allegations of academic misconduct and journals assessing claims of prior disclosure. By establishing the temporal ordering of deposits, which is frequently the disputed fact in such cases, the framework offers evidence that conventional repositories cannot readily supply, since their timestamps reside in systems that the institution itself administers.

5.3 Combatting link rot and reference rot

Zittrain *et al.* (2014) found that more than 70% of the URLs cited in the Harvard Law Review and comparable journals, and approximately 50% of those cited in United States Supreme Court opinions, no longer resolved to the material originally referenced. This failure takes two distinct forms. Link rot occurs when an address returns nothing at all, while reference rot occurs when an address returns content other than that which was cited. Addressing schemes based on location are vulnerable to both.

Content addressing eliminates the second form entirely. Because a CID is derived from the content itself, an identifier cannot resolve to material other than that from which it was computed; should the content change, the identifier changes with it. When combined with a hash recorded at the point of citation, this allows a reader to establish not merely that a resource remains available, but that it is the very resource the author consulted. For libraries operating citation support services or archiving the web resources cited in institutional output, this offers a durable response to a problem the profession has recognized for more than a decade. The considerations regarding pinning discussed in Section 6.4 apply here.

5.4 Electronic theses and dissertations

Electronic theses and dissertations represent a particularly favorable starting point. They are deposited through a workflow the library already controls, at a known moment, in a moderate annual volume, by a depositor who holds an institutional identity and has a clear and lasting interest in demonstrating the fact of deposit. The submission interface can compute the hash without requiring any change in depositor behavior, which makes this an appropriate first application for an institution wishing to evaluate the framework in practice.

5.5 Verifiable credentials and academic credentialing

Institutions can issue degrees, certificates and transcripts as verifiable records, thereby reducing both the verification burden placed on employers and institutional exposure to credential fraud. The Blockcerts deployments discussed in Section 2.1 demonstrate that this is achievable at institutional scale. Where verification currently depends upon direct correspondence with the issuing institution, the reduction in administrative effort is substantial. Where the library operates the institutional repository, it becomes a natural custodian of this infrastructure alongside the office of the registrar.

5.6 Research data and reproducibility

Sealing datasets and analysis code at the moment of publication allows a subsequent reader to establish that the data examined is precisely the data upon which the reported results were computed. This addresses a well-recognized weakness in reproducibility practice, where data are frequently deposited but rarely fixed, and it aligns closely with the principle that research data should remain retrievable through a persistent identifier. Research data services within academic libraries provide the natural home for this function.

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5.7 Legal, governmental and community records

Depository libraries, land record offices and civil registries hold documents whose evidentiary value depends entirely upon their being demonstrably unaltered. Community archives documenting displacement, land claims or human rights violations face the same requirement under considerably more difficult conditions, including the possibility that the holding institution may itself come under pressure to revise its holdings. A verification mechanism that does not depend upon the continued good faith of the custodian carries particular value in such settings. This application also carries the greatest legal complexity of those discussed here and should follow rather than precede the resolution of the governance questions raised in Section 6.

Seamless integration with existing Integrated Library Management System (ILMS) or digital repositories allows libraries to evolve into trusted digital certifiers, safeguarding the originality and legitimacy of knowledge in the digital age.

6. Challenges, limitations and implementation considerations

Despite its promising potential, implementing the CryptoBook system in libraries faces several challenges and limitations. Technical barriers, such as high blockchain gas fees and limited decentralized storage capacity, can hinder scalability and accessibility. Additionally, digital literacy gaps among library staff and users may affect the smooth adoption and management of blockchain-based tools. Legal and ethical concerns around data ownership, jurisdiction and accountability remain unresolved in many regions. Furthermore, the storage of sensitive metadata on public blockchains raises privacy issues, especially when dealing with personal or academic records.

6.1 Transaction costs

Because the prototype was executed within a simulated environment, no gas consumption was directly observed, and the figures presented here are therefore modeled rather than measured. Minting an ERC-721 token with URI storage typically consumes in the region of 90,000–150,000 gas units, depending upon metadata length and contract optimization. Taking 120,000 units as a central estimate, minting on the Ethereum Mainnet would cost in the region of one to ten United States dollars per item at prevailing gas and token prices, which for a collection of ten thousand items amounts to somewhere between ten thousand and one hundred thousand dollars. On a Layer 2 network such as Polygon, Arbitrum or Base, the same operation costs fractions of a cent, bringing the same collection within a range of tens to a few hundred dollars. On a permissioned consortium ledger, there is no transaction fee at all, and cost consists instead of infrastructure and governance overheads.

These figures carry an important implication for practitioners. Minting an entire institutional repository on the Ethereum Mainnet is unlikely to be economically defensible for most libraries at any plausible gas price, whereas the differential between mainnet and either a Layer 2 network or a permissioned ledger spans several orders of magnitude. It is therefore the choice of deployment target, rather than the choice of technology itself, that determines whether the framework is affordable. Obtaining measured rather than modeled figures, by deploying the same contract to a public test network and to a Layer 2 network, represents the most immediate priority for further work.

6.2 Key management and loss

Should a depositor lose the private key associated with a deposit, the provenance record itself survives, but the capacity to transfer or assert control over the record does not. For a student who deposits a thesis and subsequently graduates, the retention of a key across several decades is an unrealistic expectation. Several mitigations are available, each with its own difficulties. Institutional custody of keys on behalf of depositors reintroduces precisely the central authority the framework seeks to avoid; multi-signature arrangements requiring both depositor

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and institution distribute the risk but complicate the workflow; and social recovery mechanisms remain relatively untested in an archival context. This question merits careful attention before deployment rather than after it.

6.3 Integration with existing library management systems

In DSpace, the natural point of integration is a curation task or submission-step plugin invoked during the deposit workflow, with the hash and CID written to a metadata field within the item record. Islandora and Fedora expose comparable hooks through their microservices architecture. Commercial platforms such as Alma, and systems such as Koha, would require integration mediated through their application programming interfaces or the cooperation of the vendor, which is a matter of procurement rather than of technology and is likely to prove the binding constraint in practice.

One design question remains open. When a deposited item is subsequently revised, its hash necessarily changes. Whether the revision should mint a new token, update the existing record or create an explicitly linked successor is a decision with consequences for how version histories are represented and how citations to superseded versions resolve. The present framework does not settle this question, and any production implementation would need to address it.

6.4 Persistence of decentralized storage

IPFS guarantees that content is addressed by its hash, but it does not guarantee that the content remains available, since availability depends upon at least one node continuing to pin it. In the present prototype, pinning was provided by Pinata, a commercial service so that the claim of decentralized permanence rests in part upon a centralized subscription. Several options reduce this dependency, including incentivized persistence networks such as Filecoin or Arweave, institutional IPFS nodes operated by library consortia and conventional preservation storage in which content addressing serves purely as an identifier scheme. The last of these aligns most closely with existing preservation practice and would be the least disruptive for libraries to adopt.

6.5 User adoption

No user study was undertaken within the scope of this work, and questions of acceptance, workflow fit and staff capability are therefore treated here as design considerations rather than as findings. Empirical investigation drawing upon the constructs of the diffusion of innovations framework (Rogers, 2003) or an established model of technology acceptance would be required to determine how library staff and depositors respond to the workflow in practice, and this is identified as a direction for future research in Section 8.

6.6 Environmental considerations

The environmental critique of blockchain has shaped professional attitudes within the library community and is frequently the first objection raised when such proposals are considered. It is substantially weakened in respect of the proof-of-stake networks relevant here, since following the Ethereum transition of September 2022 the electricity consumption of the network fell by more than 99.9% (Crypto Carbon Ratings Institute, 2022). Permissioned ledgers consume less still. The objection retains its force against proof-of-work systems, but it does not apply to the architecture proposed here, and this distinction deserves to be made explicit whenever such proposals are brought before library governance bodies.

Addressing these challenges is essential to ensure the responsible, equitable and sustainable integration of CryptoBooks in library systems.

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7. From point A to point B: A roadmap for deployment

7.1 The idealized end state

Following the idealized design method (Ackoff *et al.*, 2006), the desired end state is first described as though it had already been achieved, and the path toward it is then derived by reasoning backwards. In such a state, every item deposited in an institutional repository carries a cryptographic seal computed at the moment of deposit, applied automatically by the submission workflow and requiring neither action nor understanding on the part of the depositor. Verification becomes an action a reader performs directly rather than a service request submitted to library staff. Seals issued by one institution are recognized and verifiable at any other within the network, much as bibliographic records are today. Provenance becomes queryable across institutional boundaries, and a citation to a sealed item resolves reliably to the item as it was cited. In this setting, the role of the librarian shifts from asserting the authenticity of holdings to operating the infrastructure through which others may establish it for themselves.

7.2 Point A: present conditions

A credible path requires an equally clear account of the present position. Most academic libraries operate DSpace, Islandora, EPrints or a commercial platform, with preservation practice built upon checksums, format migration and fixity checking. Fixity checking already employs cryptographic hashes, which means that the integrity layer of the proposed framework extends something libraries do routinely, exposing it publicly rather than holding it internally. Very few libraries currently operate wallet infrastructure or employ staff with distributed ledger expertise, and few maintain a budget line capable of absorbing per-item transaction costs. A number of institutions adopted policies discouraging engagement with distributed ledger technology following the market events of 2022 and the environmental critique, and those policies frequently remain in force. Any roadmap that fails to account for these conditions is unlikely to survive its first governance committee.

7.3 Signposts

Four signposts are proposed below. Each describes what occurs at that stage and what must already be in place for it to be reached. They indicate a sequence rather than a schedule, and the pace at which any institution moves between them will depend upon its own circumstances.

Signpost 1: Hashing-only pilots: Libraries compute SHA-512 hashes for deposited items and publish them as a metadata field within the existing repository record. No ledger, no wallet, no token and no expenditure are involved at this stage. In the terms proposed by Rogers (2003), such a pilot offers high trialability and low complexity, and it is fully reversible, requiring neither procurement nor governance approval beyond the repository manager. Its principal value lies in building the staff competence upon which every subsequent stage depends. Preconditions: none.

Signpost 2: Permissioned consortium ledgers: Groups of libraries operate a shared permissioned ledger, following the model discussed in Section 2.1, upon which hashes and depositor identifiers are recorded. Transaction costs disappear, sensitive metadata is not exposed publicly, and governance remains with the member institutions. At this point, the framework becomes operational rather than experimental, and it is reachable by existing library consortia working through their established governance structures. Preconditions: the competence developed at Signpost 1, a consortium willing to host the infrastructure and an agreed governance model covering membership, key custody and exit.

Signpost 3: Standards alignment and interoperability: The seal is expressed through existing standards rather than alongside them, with hashes and identifiers recorded in PREservation Metadata: Implementation Strategies (PREMIS) preservation metadata, bound to Digital Object Identifier (DOI), Handle or Archival Resource Key (ARK) persistent identifiers, and exposed through Open Archives Initiative Protocol for Metadata Harvesting

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(OAI-PMH). This stage determines whether the approach survives beyond its early adopters. Libraries adopt standards rather than products, and an implementation that cannot be expressed within the established metadata vocabularies of the profession is unlikely to outlast the individuals who built it. Preconditions: at least one working consortium implementation and sustained engagement with the relevant standards bodies.

Signpost 4: Cross-institutional and national integration: Seals issued by one consortium are recognized and verifiable by another so that a reader at one institution may verify a record deposited at a second. Beyond this, cryptographic sealing is incorporated into national digital preservation strategy and statutory deposit arrangements, and the evidentiary standing of a seal is clarified within the relevant jurisdiction. The organizational precedent is the shared bibliographic trust that union catalogs achieved over several decades; the difficulty is of a comparable order and is organizational rather than technical. Preconditions: standards alignment, agreements between consortia and legal analysis of evidentiary standing.

7.4 Working backwards

Read in reverse, this sequence identifies the immediate agenda with some clarity. National integration depends upon cross-institutional operation, which depends upon standards alignment, which depends upon at least one working consortium implementation, which in turn depends upon institutions possessing staff who understand the mechanism. That final condition is satisfied by Signpost 1, which costs nothing and may begin immediately. The binding constraint upon the entire sequence is therefore institutional competence rather than cost, cryptography or vendor support, and the first signpost is designed specifically to build it. Three questions warrant resolution before Signpost 2 can reasonably be attempted: what custody arrangement discharges the institutional duty of care toward depositors who cannot be expected to retain keys across decades; how revisions and retractions may be represented within an append-only record without orphaning citations to superseded versions; and what obligation a member institution carries to maintain verifiability should a consortium subsequently dissolve.

8. Conclusions

The CryptoBook framework offers a transformative solution for libraries to authenticate, preserve and verify digital documents with long-term reliability. By integrating SHA-512 hashing with blockchain-based NFTs, libraries can ensure tamper-evident records, decentralized ownership and trustless verification. This system empowers users to independently validate document authenticity and authorship, promoting transparency and accountability. Despite challenges such as technical barriers, metadata privacy concerns and limited digital literacy, the model remains accessible through Python scripting, IPFS storage and no-code NFT platforms. As digital collections grow and trust in information becomes critical, CryptoBooks provides libraries with a forward-looking tool that enables them to become not just repositories of knowledge but verifiable custodians of truth.

Beyond the technical demonstration, the study locates the framework within established archival theory, showing that its three components correspond to the three conditions under which a record may be presumed authentic. It also identifies seven areas of practical application and sets out a staged roadmap whose first stage requires no ledger, no expenditure and no institutional risk.

Four directions for future work follow. The most immediate is deployment of the same contract to a public test network and to a Layer 2 network, replacing the modeled cost figures of Section 6.1 with measured values. The second is a pilot at Signpost 1, which would generate the first evidence on workflow fit and staff capability. The third is an empirical study of adoption among library staff and depositors. The fourth is legal analysis of the evidentiary status of a cryptographic seal in the relevant jurisdiction.

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From a policy perspective, adopting CryptoBooks can support institutional repositories, digital credentialing systems and national digital preservation strategies by providing verifiable, tamper-resistant records. This positions libraries as critical infrastructures for trust and authenticity in the digital knowledge ecosystem.

Ethics and consent statement

This study did not involve human participants, personal data or animal subjects. It comprises the design and technical demonstration of a software artifact. Ethical approval and participant consent were therefore not required.

Declaration of generative AI use

The authors used a generative artificial intelligence tool solely for grammatical correction and language editing of the manuscript. It was not used in the study design, data collection, statistical analysis, interpretation, sourcing or verification of references or in any scholarly judgment reported in this article. No artificial intelligence tool is listed as an author. The authors take full responsibility for the content, accuracy and integrity of this work.

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